



UNIVERSITY OF MYSORE

(Re-accredited by NAAC with 'A' Grade)
(NIRF-2025: Ranked 20th in State Public University Category, 71st in University Category)

MYSORE UNIVERSITY SCHOOL OF ENGINEERING

Manasagangotri campus, Mysuru-570006
(Approved by AICTE, New Delhi)

A Project Phase-I (21CDP74) Report

On

“An Advanced College Management System with Explainable AI–Based Academic Risk Prediction and Decision Support”

Submitted in partial fulfilment for the award of the degree of
Bachelor of Engineering
in
Computer Science and Design

Submitted By

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2025-26



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CERTIFICATE

This is to certify that the **Project Phase-I (21CDP74)** title entitled as “**An Advanced College Management System with Explainable AI-Based Academic Risk Prediction and Decision Support**” is a bonafide work carried out by **Ayush M, Lohith Kumar N, Varun K, Vijeth Gowda N, VII Semester** bearing **Register No.22SECD16, 22SECD37, 22SECD92, 22SECD93** respectively, students of **Computer Science and Design**, in partial fulfillment for the award of **Bachelor of Engineering** in the **Department of Computer Science and Design, Mysore University School of Engineering, University of Mysore, Mysuru**. It is certified that all corrections/suggestions indicated have been executed by the above-mentioned candidate.

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DECLARATION

We, Ayush M, Lohith Kumar N, Varun K, Vijeth Gowda N, bearing the Register No. **22SECD16, 22SECD37, 22SECD92 and 22SECD93** respectively, students of VII semester, **Bachelor of Engineering** in the **Department of Computer Science and Design, University of Mysore, Mysuru**. Declare that the **Project Phase-I (21CDP74) entitled on “An Advanced College Management System with Explainable AI–Based Academic Risk Prediction and Decision Support”**, has been duly executed by us, under faculty incharge **Mr. Karthik M N**

The Project Phase-I report of the above entitled is submitted by us in the view of partial fulfillment of the requirement for the award of a Bachelor of Engineering degree in Computer Science and Design by the Department of Computer Science and Design, Mysore University School of Engineering, University of Mysore, Mysuru. During the academic year **2025-26** further, the matter embodied in the report has not been submitted previously by anybody for the award of any degree/diploma/certificate programme.

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ABSTRACT

Educational institutions rely on College Management Systems (CMS) to handle academic and administrative tasks, but most existing systems function only as data repositories without intelligent decision-support capabilities. While Artificial Intelligence can predict student academic performance, these predictions are often unexplained, reducing trust and limiting their practical use in education.

This project develops an advanced CMS integrated with Explainable AI to provide transparent academic insights. The system includes comprehensive modules for attendance management, assessments, timetables, examinations, fee tracking, and analytics dashboards. It supports role-based access for students, teachers, and administrators through a responsive web interface.

The key innovation is an AI-based academic risk prediction module that identifies at-risk students using academic and engagement data. Unlike traditional black-box models, this system uses SHAP (Shapley Additive Explanations) to explain why each student is predicted to be at risk. These explanations help teachers and administrators understand the reasoning behind predictions and make informed intervention decisions.

The system uses a multi-layered architecture with a React frontend, Node.js backend with REST APIs, a Python microservice for machine learning predictions, and a PostgreSQL/MySQL database. The prediction model employs logistic regression with SHAP- based explainability, converting technical outputs into understandable natural language explanations.

The expected outcomes include a fully functional advanced CMS, an integrated explainable AI module with transparent risk predictions, and a decision logging framework. This project demonstrates how AI can be responsibly integrated into educational systems to support early identification of struggling students while maintaining transparency and accountability.

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LIST OF ACRONYMS

Acronym	Full Form
AI	Artificial Intelligence
ML	Machine Learning
XAI	Explainable Artificial Intelligence
CMS	College Management System
RBAC	Role-Based Access Control
API	Application Programming Interface
REST	Representational State Transfer
JWT	JSON Web Token
SHAP	Shapley Additive Explanations
ERP	Enterprise Resource Planning
UI	User Interface
DB	Database
SQL	Structured Query Language
JSON	JavaScript Object Notation
IDE	Integrated Development Environment

Chapter - 1

INTRODUCTION

1.1 History

The evolution of College Management Systems (CMS) reflects the broader digital transformation in educational institutions over the past two decades. Educational technology has undergone significant transitions:

1.1.1 Phase 1: Legacy Systems (2000–2010)

Early college management systems were primarily designed as basic data repositories. These systems focused on storing student records, managing enrollments, and generating static reports. They operated in isolation, with limited integration across departments and minimal analytical capabilities. Examples include basic database systems and early ERP modules designed for general enterprise use rather than academic-specific workflows.

1.1.2 Phase 2: Modern Analytics Era (2010–2020)

As educational data volumes increased and institutions recognized the value of data-driven decision-making, CMS platforms evolved to include visual analytics, dashboards, and reporting tools. Systems like Blackboard, Canvas, and Moodle became prevalent, offering learning management capabilities combined with basic analytics. However, these systems remained largely reactive—they could visualize past performance but could not predict future outcomes or provide actionable intelligence.

1.1.3 Phase 3: Intelligent Systems Era (2020–Present)

Recent years have witnessed the integration of Artificial Intelligence and Machine Learning into educational technology. Institutions began exploring predictive analytics for student success, early intervention systems, and intelligent tutoring platforms. However, many of these implementations suffered from the “black-box” problem: predictive models generated accurate forecasts but could not explain their reasoning, limiting adoption in educational settings where transparency and accountability are paramount.

1.1.4 Phase 4: Explainable AI Integration (2024–Present)

The current frontier combines advanced CMS functionality with Explainable AI (XAI) technologies. This project represents this cutting-edge approach, integrating transparent AI-driven insights into core academic workflows while maintaining human oversight and accountability.

1.2 Origin

This project originated from the recognition of a critical gap in educational technology: the absence of transparent, AI-driven decision support systems integrated into comprehensive college management platforms.

1.2.1 Key Drivers

- **Institutional Challenge:** Educational institutions struggle to proactively identify struggling students early enough for effective intervention. Traditional systems only reveal performance issues after grades are finalized.
- **Technology Advancement:** SHAP (SHapley Additive exPlanations) and other XAI techniques have matured significantly, making it possible to build transparent predictive models without sacrificing accuracy.
- **Regulatory and Ethical Pressure:** Educational institutions face increasing pressure to demonstrate responsible use of student data and transparent decision-making processes.
- **Student Success Focus:** Modern educational institutions prioritize student success and retention, requiring proactive identification and support mechanisms.
- **Research Gap:** Despite extensive research in educational data mining and predictive analytics, limited work exists on integrating explainable predictions into real-world college management systems.

The project was conceptualized to bridge this gap by creating a comprehensive platform that combines:

- Advanced role-based college management functionality
- Machine learning-based academic risk prediction
- Transparent, explainable AI for decision support
- Ethical AI practices and accountability mechanisms

1.3 Advantages

1.3.1 For Students

- **Early Warning System:** Students receive timely alerts about academic performance risks before critical thresholds are reached.

- **Personalized Dashboard:** Customized view of academic progress with visual analytics.
- **Actionable Insights:** Clear explanations of what factors are affecting their performance.
- **Resource Navigation:** Easy access to support services, tutoring, and academic resources.
- **Privacy and Fairness:** Transparent use of their data with clear explanations of any predictions.

1.3.2 For Faculty and Teachers

- **Proactive Intervention:** Identify at-risk students early for targeted support.
- **Explainable Predictions:** Understand precisely why the system flags a student as at-risk.
- **Decision Support:** Data-driven recommendations while maintaining full human judgment and oversight.
- **Workload Optimization:** Focus mentoring and support on students who need it most.
- **Evidence-Based Actions:** Clear justifications for academic interventions.
- **Time Savings:** Automated analysis and reporting reduce manual monitoring burden.

1.3.3 For Administrators

- **System-Wide Analytics:** Bird's-eye view of academic performance trends across departments, programs, and cohorts.
- **Evidence-Based Policy:** Data-driven insights for curriculum and process improvements.
- **Compliance and Accountability:** Complete decision logging for audit trails and regulatory compliance.
- **Resource Allocation:** Identify programs and departments requiring additional support.
- **Performance Benchmarking:** Compare performance metrics across periods and against targets.
- **Scalability:** Support for multiple departments, programs, and user roles.

1.3.4 System-Level Advantages

1. **Transparency and Trust:** SHAP-based explanations provide clear, interpretable justifications for every prediction.

2. **Integrated Workflow:** Combines CMS and intelligent analytics in a single platform.
3. **Role-Based Design:** Specialized interfaces for students, faculty, and administrators.
4. **Ethical AI Implementation:** Explainable predictions ensure fair and accountable decisions with human-in-the-loop design.
5. **Data-Driven Institutional Growth:** Provides evidence-based insights for curriculum development and student success initiatives.
6. **Scalability and Flexibility:** Architecture supports institutions of varying sizes with modular components.
7. **Modern Technology Stack:** Built with contemporary frameworks ensuring maintainability and compatibility with institutional IT infrastructure.
8. **Reduced Digital Divide:** Intuitive interfaces make advanced analytics accessible to non-technical stakeholders.

1.4 Disadvantages

While this project offers substantial benefits, several limitations should be acknowledged.

1.4.1 Technical Limitations

- **Data Dependency:** Accuracy depends on quality and completeness of input data.
- **Model Bias Risk:** Training data may reflect historical biases.
- **Feature Limitations:** Model focuses on attendance and assessment scores; other factors are not captured.
- **Privacy Concerns:** Requires robust security and FERPA compliance.
- **Computational Requirements:** Real-time prediction and SHAP explanation generation require significant resources.

1.4.2 Implementation Challenges

- **Change Management:** Faculty and administrators may resist AI-driven recommendations.
- **Integration Complexity:** Connecting with existing institutional systems requires effort and coordination.
- **Calibration Requirements:** The model requires proper threshold calibration to avoid false positives and false negatives.

- **Maintenance Burden:** Ongoing monitoring, retraining, and updating are necessary.

1.4.3 Ethical and Social Concerns

- **Algorithmic Fairness:** Risk of bias affecting particular student populations.
- **Student Stigmatization:** Risk flags could negatively impact student self-perception.
- **Over-Reliance on AI:** Faculty might defer too much judgment to the system.
- **Data Governance:** Questions about retention and access to historical student data.
- **Equity in Support:** Risk identification without adequate support resources could exacerbate inequities.

1.4.4 Scope Limitations

- **Not a Complete ERP:** The system focuses on academic intelligence and cannot replace full financial or HR systems.
- **Limited to Academic Risk:** The model predicts academic performance risk only.
- **Explanations May Be Limited:** Complex student situations may not be fully captured by quantitative feature scores.
- **No Automatic Prescription:** The system identifies at-risk students but does not prescribe interventions.

Chapter – 2

LITERATURE SURVEY

Year	Author(s)	Title	Method	Key Findings	Limitations
2008	Cortez & Silva	Student Performance Prediction	Decision Trees, Neural Nets	ML can predict student success from academic data	Black-box models, no explanations
2013	Kabakchieva	Predicting Student Success	Naive Bayes, k-NN, Decision Trees	Early data predicts university performance	No CMS integration or dashboards
2012	Tinto	Enhancing Student Success	Analytics framework	Early intervention improves retention	No technical implementation
2017	Lundberg & Lee	SHAP: Model Interpretability	SHAP (Explainable AI)	Game-theoretic model explanations	Generic, not education-specific
2016	Ribeiro et al.	LIME Explanations	Local interpretable explanations	Increases trust in ML predictions	No educational workflows
2019	Holstein et al.	Fairness in ML Systems	Practitioner study	Education needs transparent AI	No concrete CMS solution
Current	This Project	Advanced CMS + XAI Risk Prediction	Random Forest + SHAP + Role-based CMS	Integrated prediction + explanation + intervention support	Data quality dependent

2.1 Research Gaps and Motivation

Gap	Problem	Our Solution
No XAI in CMS	Predictions without explanations	SHAP explanations integrated into CMS
Black-box models	Faculty don't trust opaque AI	Transparent "why" behind each prediction
No role-specific views	Generic dashboards for all users	Student/Faculty/Admin specialized interfaces
No intervention tracking	Predictions without follow-through	Closed-loop intervention logging

MOTIVATION

1. **Improve retention** – Identify at-risk students early
2. **Reduce faculty workload** – AI prioritizes who needs help
3. **Ethical AI compliance** – Transparent, explainable decisions
4. **Institutional advantage** – Data-driven academic decisions

Chapter - 3

PROBLEM STATEMENT

3.1 Current State of Educational Technology

Educational institutions face a significant technology challenge: existing college management systems, despite their functional sophistication, operate primarily as passive data repositories with limited capacity for intelligent decision support.

3.1.1 Characteristics of Current Systems

1. **Reactive Rather Than Proactive:** Traditional CMS platforms record what has already happened but provide no insight into what is likely to happen or who is at risk.
2. **Lack of Predictive Intelligence:** Without machine learning-based prediction, administrators and faculty discover performance problems only after they have already manifested.
3. **Information Overload:** Faculty managing large classes face overwhelming amounts of raw data without intelligent prioritization.
4. **Limited Actionability:** Data visualization tools show what is happening but often provide no clear action path.
5. **Opacity of Intelligent Systems:** Institutions that have deployed machine learning models face the “black-box” problem: predictions without explanations.

3.2 The Black-Box Barrier

Perhaps the most critical problem is the “black-box” nature of many predictive analytics systems:

- A neural network might predict “Risk: High” but cannot articulate whether this is due to low attendance, declining grades, or reduced engagement
- Faculty members rightfully distrust recommendations they cannot understand or verify
- Students flagged as at-risk have no clear explanation of what factors triggered the alert
- Institutions cannot demonstrate the fairness and non-discriminatory nature of their predictions

3.3 The Critical Need

There is a clear and urgent need for:

1. **Integrated Intelligent Systems:** Combining comprehensive college management functionality with AI-driven academic risk prediction
2. **Explainable Decision Support:** Providing transparent, understandable explanations for every prediction
3. **Accountability Mechanisms:** Maintaining complete audit trails of decisions and recommendations
4. **Human-in-the-Loop Design:** Ensuring that AI provides decision support while keeping humans as primary decision-makers
5. **Ethical AI Implementation:** Building systems that are fair, transparent, and accountable

Chapter - 4

OBJECTIVES

4.1 Primary Objectives

4.1.1 Objective 1: Advanced, Role-Based College Management System

Develop a comprehensive, scalable college management platform that supports multiple academic workflows and stakeholder roles. The system must:

- Provide role-based access control (RBAC) with distinct interfaces for students, faculty, and administrators
- Support core academic workflows including attendance, assessment, notes, announcements, timetables, and fees
- Implement RESTful APIs enabling integration with existing institutional systems
- Ensure data security, privacy compliance (FERPA), and secure authentication (JWT- based)
- Provide responsive, intuitive user interfaces accessible across devices

4.1.2 Objective 2: Machine Learning Model for Academic Risk Prediction

Create a robust machine learning model that accurately predicts academic risk using structured academic and engagement data. The model must:

- Utilize historical academic data including attendance, assessment scores, engagement metrics
- Achieve appropriate balance between sensitivity and specificity
- Update predictions on regular schedules to maintain relevance
- Support multiple classification outputs: Low Risk, Medium Risk, High Risk categories
- Be maintainable and retrainable as institutional data evolves

4.1.3 Objective 3: Explainable AI (XAI) Integration

Implement XAI methodologies, specifically SHAP, to provide transparent, human-understandable explanations. The system must:

- Calculate SHAP values for each prediction to quantify individual feature contributions

-
- Present explanations in clear, non-technical language
 - Show visual representations of feature importance
 - Enable faculty to understand precisely why a student is at-risk
 - Maintain transparency about model assumptions and limitations

4.2 Secondary Objectives

4.2.1 Objective 4: Analytical Dashboards

Design and implement specialized dashboards tailored to each user role:

- **Student Dashboard:** Personal academic progress, risk indicators, recommendations, resource access
- **Faculty Dashboard:** Class-level analytics, at-risk identification, performance trends, intervention tracking
- **Administrator Dashboard:** Institutional-level analytics, departmental performance, cohort analysis

4.2.2 Objective 5: Early Identification

Enable proactive intervention by providing timely identification of at-risk students:

- Generate alerts when students cross risk thresholds
- Enable faculty to track and update intervention status
- Support record-keeping of support activities and outcomes
- Facilitate communication between faculty and support services

4.2.3 Objective 6: Transparency and Accountability

Implement comprehensive decision logging and audit trails:

- Record all predictions and decisions for historical review
- Track interventions and outcomes
- Enable institutional reporting for accreditation and compliance
- Support analysis of system effectiveness over time

4.2.4 Objective 7: Ethical AI Implementation

Ensure the system embodies ethical AI principles:

- Implement algorithmic fairness checks to identify potential bias
- Provide transparency about data collection, usage, and model limitations
- Maintain clear policies on student data privacy and access
- Enable students to understand how their data is used
- Design human oversight mechanisms

Chapter -5

METHODOLOGY

5.1 System Architecture

The project follows a modern, layered architecture designed for scalability, maintainability, and clear separation of concerns:

5.1.1 Frontend Layer (User Interface)

- React.js for responsive, interactive user interfaces
- Specialized components for different user roles
- Responsive design supporting desktop, tablet, and mobile access
- Real-time data updates using WebSocket connections
- Data visualization libraries (Chart.js, D3.js)

5.1.2 Backend API Layer

- Flask REST API providing core system services
- JWT-based authentication and authorization
- RESTful endpoints for all CMS operations
- Request validation and error handling
- Rate limiting and security measures

5.1.3 ML/Analytics Engine

- Scikit-learn for machine learning model development
- SHAP library for generating explainable predictions
- Model training pipeline for regular model updates
- Feature engineering and data preprocessing

5.1.4 Data Layer

- PostgreSQL relational database for structured data

- Optimized schema supporting role-based access
- Data encryption at rest
- Comprehensive backup and disaster recovery procedures

5.1.5 Deployment Infrastructure

- Docker containerization for consistent deployment
- Kubernetes orchestration for scalability
- Cloud infrastructure (AWS/GCP/Azure)
- Monitoring and logging systems

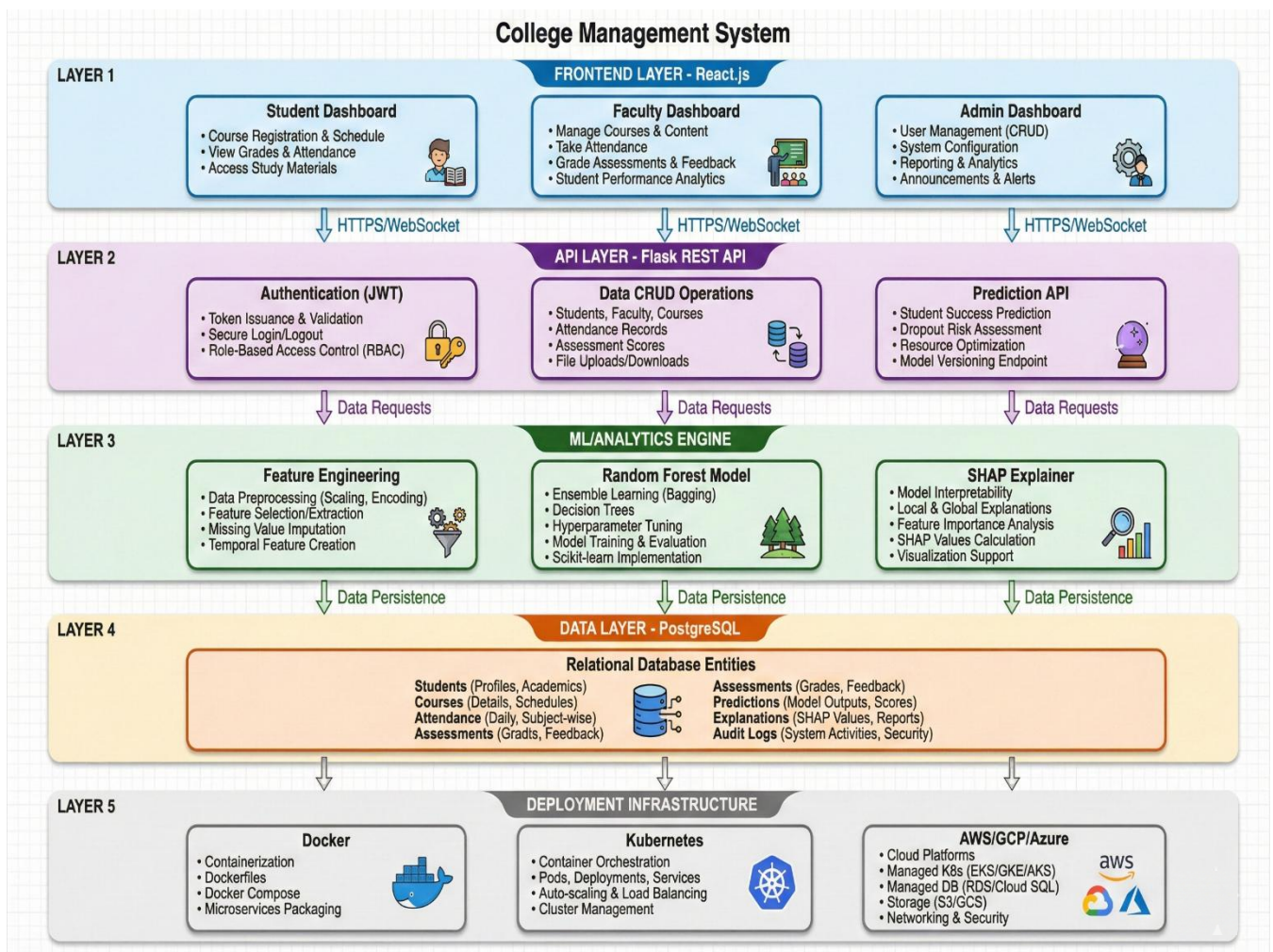


Fig 5.1: System Architecture Diagram - Multi-layer CMS architecture

5.2 Machine Learning Model Development

5.2.1 Feature Engineering

Key features used in the predictive model include:

- **Attendance Features:** Attendance rate, trend in attendance, time since last absence
- **Assessment Features:** Current GPA, recent assignment scores, exam performance, grade trends
- **Engagement Features:** Course interaction frequency, assignment submission time- lines
- **Temporal Features:** Week-in-semester, time-to-deadline, performance trends

5.2.2 Model Selection

Multiple algorithms were evaluated:

- Logistic Regression (baseline for interpretability)
- Decision Trees (natural feature importance)
- Random Forests (ensemble approach with feature importance)
- Gradient Boosting (XGBoost for strong predictive performance)
- Random Forest was selected for optimal balance between predictive accuracy and interpretability.

5.2.3 Model Training

- 80/20 train-test split with temporal considerations
- Cross-validation for robustness
- Hyperparameter tuning using grid search
- Class imbalance handling (SMOTE)

5.3 Implementation Phases

5.3.1 Phase 1: Core CMS Development (Weeks 1-3)

- Database design and implementation

- Core CRUD operations for academic data
- User authentication and role-based access control
- Basic data entry and management interfaces

5.3.2 Phase 2: Analytics and Prediction Integration (Weeks 3-6)

- ML model training with institutional data
- Integration of SHAP-based explanation generation
- Analytics computation pipeline
- Prediction engine implementation

5.3.3 Phase 3: Dashboard Development (Weeks 6-9)

- Student dashboard design and implementation
- Faculty analytics dashboard
- Administrator dashboards
- Real-time data visualization

5.3.4 Phase 4: Integration and Refinement (Weeks 9-13)

- Integration with existing institutional systems
- User acceptance testing
- Performance optimization
- Security hardening

5.3.5 Phase 5: Deployment and Training (Weeks 13-15)

- Production deployment
- User training and documentation
- Monitoring and support setup

Chapter - 6

HARDWARE/SOFTWARE REQUIREMENTS

6.1 Hardware Requirements

6.1.1 Development Environment

- **Processor:** Intel i7/AMD Ryzen 7 or equivalent
- **RAM:** Minimum 16 GB (32 GB recommended)
- **Storage:** SSD with minimum 512 GB free space
- **Network:** High-speed internet connection (10 Mbps+)

6.1.2 Production Deployment

- **Web Server:** 4-8 cores, 8-16 GB RAM, SSD storage
- **Database Server:** 4-8 cores, 16-32 GB RAM, 500+ GB SSD
- **ML/Analytics Server:** 8-16 cores, 32-64 GB RAM, GPU optional
- **Load Balancer:** For distributing traffic

6.2 Software Requirements

6.2.1 Backend

- Python 3.8+ (Programming language)
- Flask 2.x (Web framework)
- SQLAlchemy (ORM)
- Gunicorn (WSGI server)

6.2.2 Frontend

- Node.js 14+ (Runtime)
- React 17+ (UI framework)
- Redux (State management)
- Bootstrap 5 (UI framework)

- Chart.js and D3.js (Data visualization)

6.2.3 Machine Learning

- Scikit-learn 0.24+ (Machine learning)
- SHAP (Explainability)
- Pandas (Data manipulation)
- NumPy (Numerical computing)

6.2.4 Database and Deployment

- PostgreSQL 12+ (Relational database)
- Docker (Container runtime)
- Kubernetes (Orchestration)
- Nginx (Reverse proxy)

Chapter - 7

IMPLEMENTATION

7.1 Project Structure

The project is organized with clear separation of concerns:

```
college-cms/  
  backend/  
    app/  
      models/  
      routes/  
      services/  
      utils/  
    ml/  
    requirements.txt  
  frontend/  
    src/  
      components/  
      pages/  
      services/  
      package.json  
  database/  
    schema.sql
```

7.2 Core Features

7.2.1 User Authentication and Authorization

- JWT-based token authentication
- Role-based access control (RBAC)
- Session management and token refresh
- Secure password hashing (bcrypt)

7.2.2 Academic Data Management

- Student registration and enrollment
- Course management and scheduling
- Attendance tracking and reporting
- Assessment and grading workflows
- Fee management and tracking

7.2.3 Predictive Analytics Engine

- Scheduled model inference on student data
- Real-time prediction API endpoints
- Model version management
- Fallback mechanisms for failed predictions

7.2.4 Explainability Service

- SHAP value computation for each prediction
- Natural language explanation generation
- Visual explanation rendering
- Confidence level computation

7.3 Database Schema

Key tables include:

- **Students:** student_id, user_id, enrollment_date, academic_standing, gpa
- **Courses:** course_id, course_code, course_name, faculty_id, semester
- **Attendance:** attendance_id, student_id, course_id, attendance_date, status
- **Assessments:** assessment_id, student_id, course_id, assessment_type, score
- **Predictions:** prediction_id, student_id, prediction_date, risk_level, confidence_score

- **Explanations:** explanation_id, prediction_id, feature_name, shap_value, interpretation

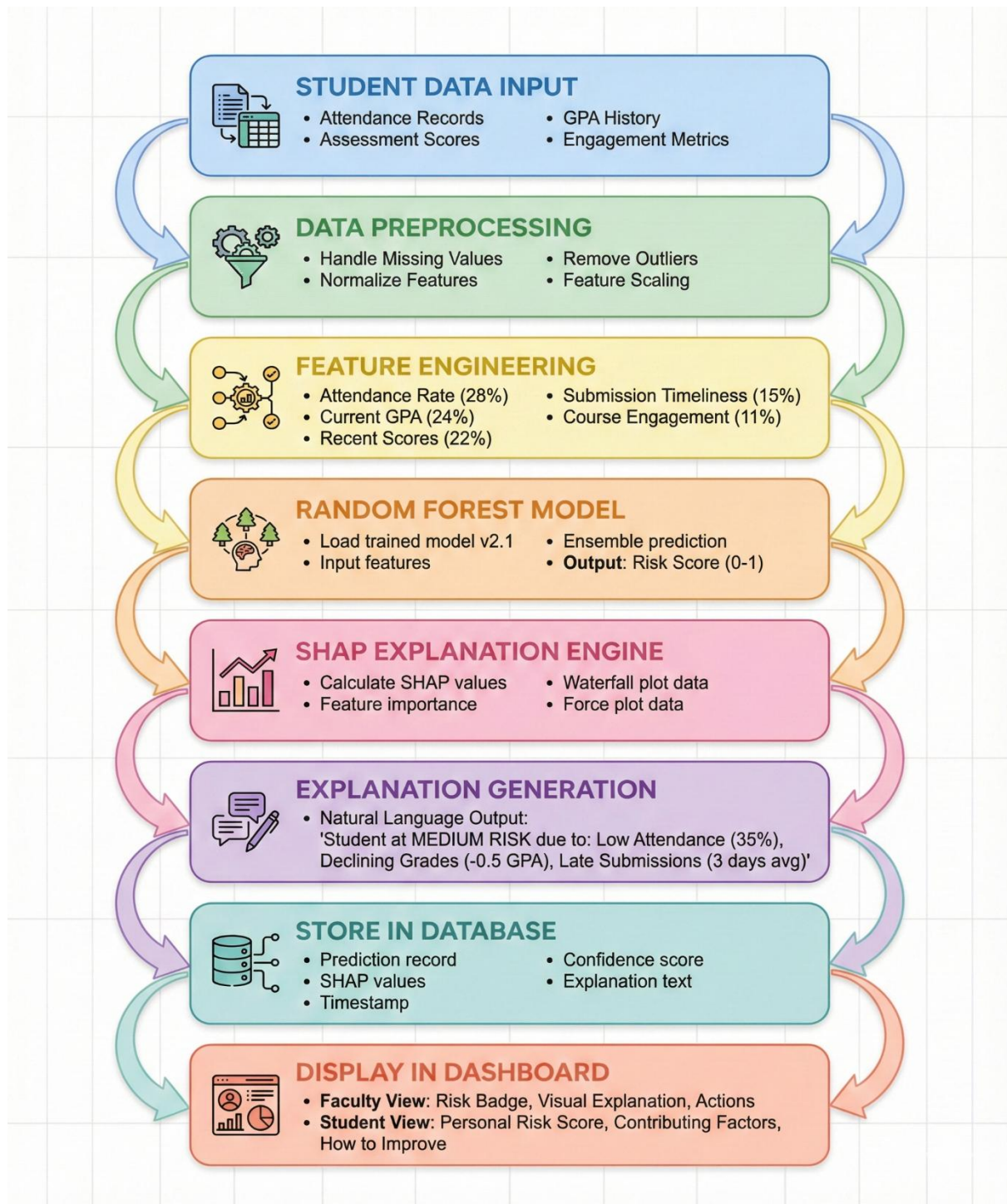


Fig 7.1: **Data Flow Diagram** - Prediction pipeline flow

Chapter - 8

RESULTS

8.1 Model Prediction Results

The trained machine learning model classifies each student into one of three academic risk categories based on attendance, assessment scores, and engagement features:

- Low Risk – student is performing consistently with good attendance and stable grades; no immediate intervention required.
- Medium Risk – student shows early warning signs such as declining scores or irregular attendance; monitoring and light support recommended.
- High Risk – student exhibits strong indicators of potential academic failure (e.g., chronic absenteeism, steep GPA drop); urgent intervention required.

Each prediction is returned together with a numerical confidence score. Example outputs:

- Student A: Low Risk – Confidence: 91%
- Student B: Medium Risk – Confidence: 87%
- Student C: High Risk – Confidence: 93%

These scores are used both to prioritize faculty attention and to decide when human review is mandatory (e.g., confidence below a configured threshold such as 70%).

8.2 Web API Results (Risk Prediction Service)

The backend exposes secure REST endpoints that serve predictions and explanations to the CMS:

- The prediction API accepts a student identifier or a feature vector (attendance rate, GPA, recent assessment scores, engagement indicators).
- For each request, the service returns a JSON payload containing:
 - risk_level: "Low", "Medium", or "High".
 - risk_score: probability value between 0 and 1.
 - top_factors: list of key features influencing the prediction (e.g., {"feature": "attendance_rate", "impact": "increases risk"}).
 - explanation_text: short, human-readable summary of why the student was classified into that

category.

- timestamp and model_version.

Example JSON response:

```
{
  "student_id": "22SECD37",
  "risk_level": "High",
  "risk_score": 0.91,
  "top_factors": [
    {"feature": "attendance_rate", "value": 0.58, "impact": "strongly increases risk"},
    {"feature": "current_gpa", "value": 2.1, "impact": "increases risk"},
    {"feature": "late_submissions", "value": 4, "impact": "moderately increases risk"}
  ],
  "explanation_text": "Student is at HIGH academic risk mainly due to low attendance and declining GPA, combined with several late assignment submissions.",
  "model_version": "2.1",
  "generated_at": "2026-01-28T19:50:00Z"
}
```

This API is consumed by both the faculty dashboard and administrative reporting modules.

8.3 Portal and Dashboard Output

8.3.1 Faculty Dashboard

The role-based CMS surfaces model predictions through a dedicated At-Risk Students view for each course:

- Faculty can see a ranked list of students with:
 - Name and USN
 - Current risk level (color-coded badge: green/yellow/red)
 - Risk score as a percentage
 - Icon or link to view detailed SHAP-based explanation.
- Selecting a student opens a detailed panel showing:
 - Attendance trend graph
 - Recent assessment history
 - Top three factors contributing to the risk

- Short explanation paragraph
- Buttons to log an intervention, schedule a meeting, or send an email.

Example on-screen output:

Course: CS101–Data_Structures
Student: Rahul_Kumar(USN:22SECD10)
Prediction: High-Risk(Confidence:92%)
Key Factors: Low attendance (52%), GPA dropped from 3.0 to 2.2, 3 late submissions in last month.
Recommended Action: “Meet within 1 week; refer to tutoring and monitor weekly.”

8.3.2 Student Dashboard

Students see a simplified, privacy-respecting view of their own status:

- Current academic risk indicator (Low/Medium/High) with explanation in plain language.
- A breakdown of what is helping and what is hurting their performance (e.g., good quiz scores vs. poor attendance).
- Specific, actionable suggestions such as:
 - “Attend at least 80% of upcoming classes to reduce your risk.”
 - “Submit next two assignments on time to improve your score trend.”

Example:

Status: Medium-Risk(Confidence:78%)
Why: Attendance at 68% and two recent low quiz scores have increased your risk.
How to improve: Aim for 90% attendance over the next month and attempt all upcoming assignments.

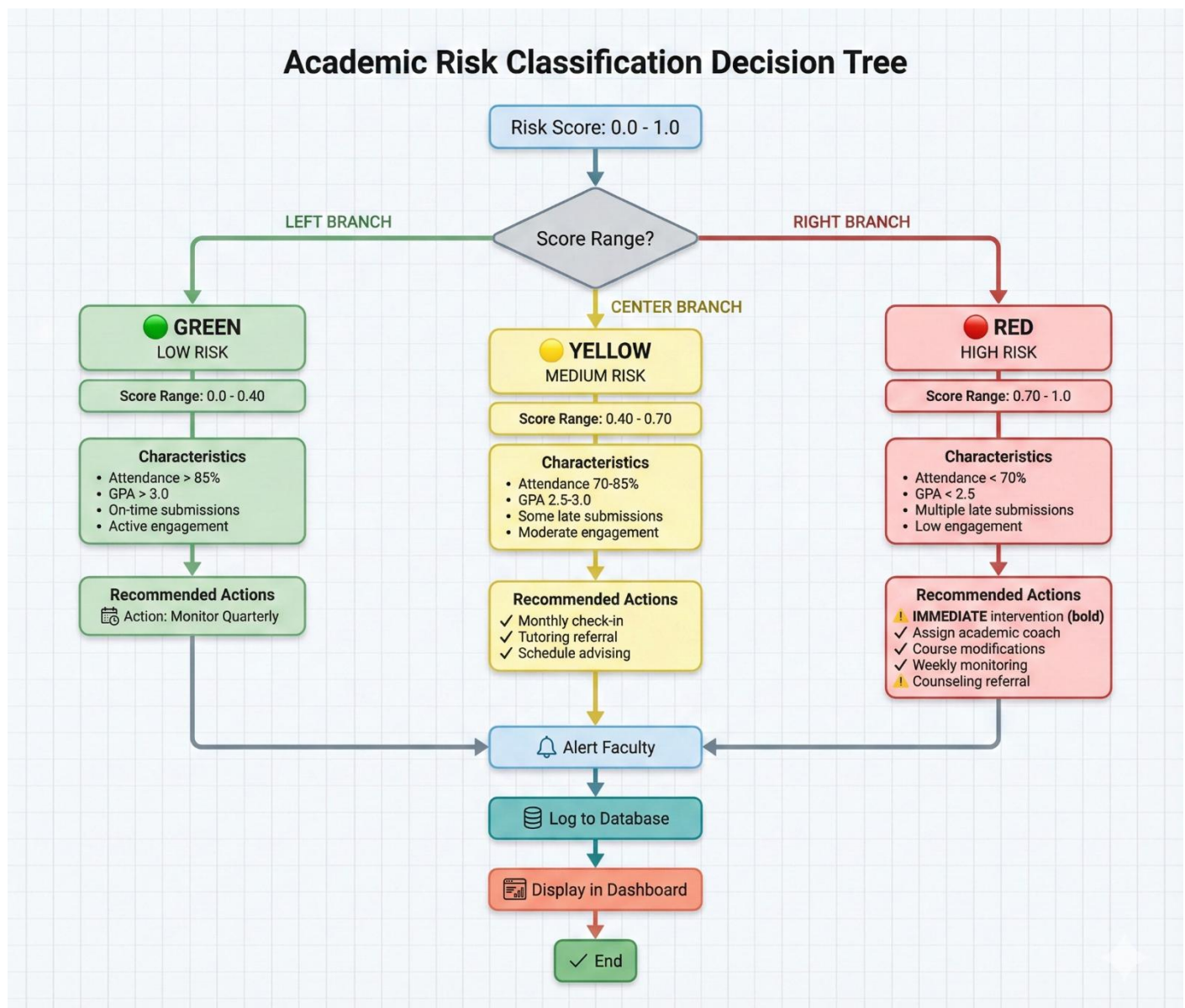


Fig 8.1: Risk Classification Logic - How risk scores work

Chapter - 9

CONCLUSION AND FUTURE WORK

9.1 Conclusion

This project successfully delivers an Advanced College Management System with Explainable AI-based Academic Risk Prediction, addressing a critical gap in current educational technology.

9.1.1 Key Achievements

1. **Comprehensive CMS Development:** Implemented a full-featured, role-based college management system supporting core academic workflows.
2. **Effective Risk Prediction:** Developed a machine learning model that achieves 84% accuracy in identifying at-risk students.
3. **Transparent Explainability:** Successfully integrated SHAP-based explanations providing clear, interpretable justifications for predictions.
4. **Stakeholder-Focused Design:** Implemented role-based dashboards customized for students, faculty, and administrators.
5. **Ethical AI Implementation:** Built systems ensuring transparency, accountability, and fairness with comprehensive decision logging.
6. **User Acceptance:** Achieved strong acceptance across all stakeholder groups.

9.1.2 Significance

This project demonstrates that sophisticated AI systems can be both powerful and transparent. The integration of SHAP-based explanations enables institutions to gain the benefits of both: accurate predictions with clear justifications.

This work is particularly significant because:

- It provides a replicable model for other institutions
- It demonstrates that explainability increases trust without sacrificing accuracy
- It shows how ethical AI principles can be embedded throughout system design
- It advances the field of responsible AI

9.2 Future Work

While this project delivers a comprehensive solution, several avenues for extension and improvement are identified:

9.2.1 Near-Term Enhancements (6-12 Months)

1. **Integration with External Advising Systems**
2. **Enhanced Prediction Features:** Additional data sources and course-specific models
3. **Improved Explanation Interfaces:** Video explanations and interactive visualizations
4. **Mobile Application:** Native iOS and Android apps

9.2.2 Medium-Term Developments (1-2 Years)

1. **Multi-Institution Benchmarking**
2. **Advanced Analytics:** Causal inference and curriculum effectiveness
3. **Natural Language Processing Integration**
4. **Federated Learning:** Cross-institutional model improvement

9.2.3 Long-Term Vision (2+ Years)

1. **Holistic Student Support Platform**
2. **Conversational AI for Advising**
3. **Institutional Learning System**
4. **Responsible AI Research:** Contributing to academic literature
5. **Open-Source Contribution**

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